

MLE4228 Sem1 Presentation

Lab4 Thermal Actuators

Group 2 Lu Jianyi

catalogue

01

Introduction and Objective

02

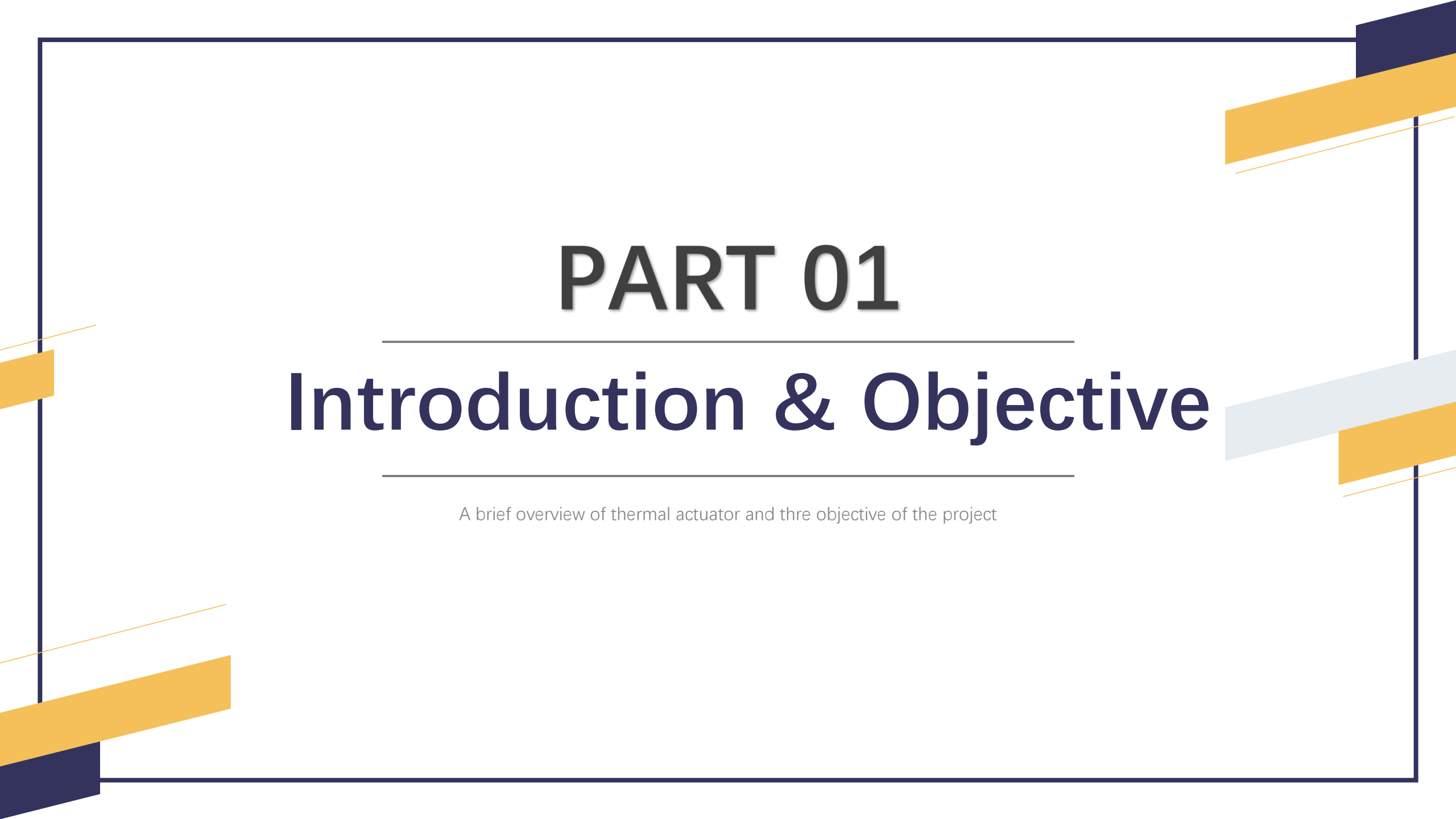
Materials and Methods

03

Results and Discussion

04

Conclusion and References



PART 01

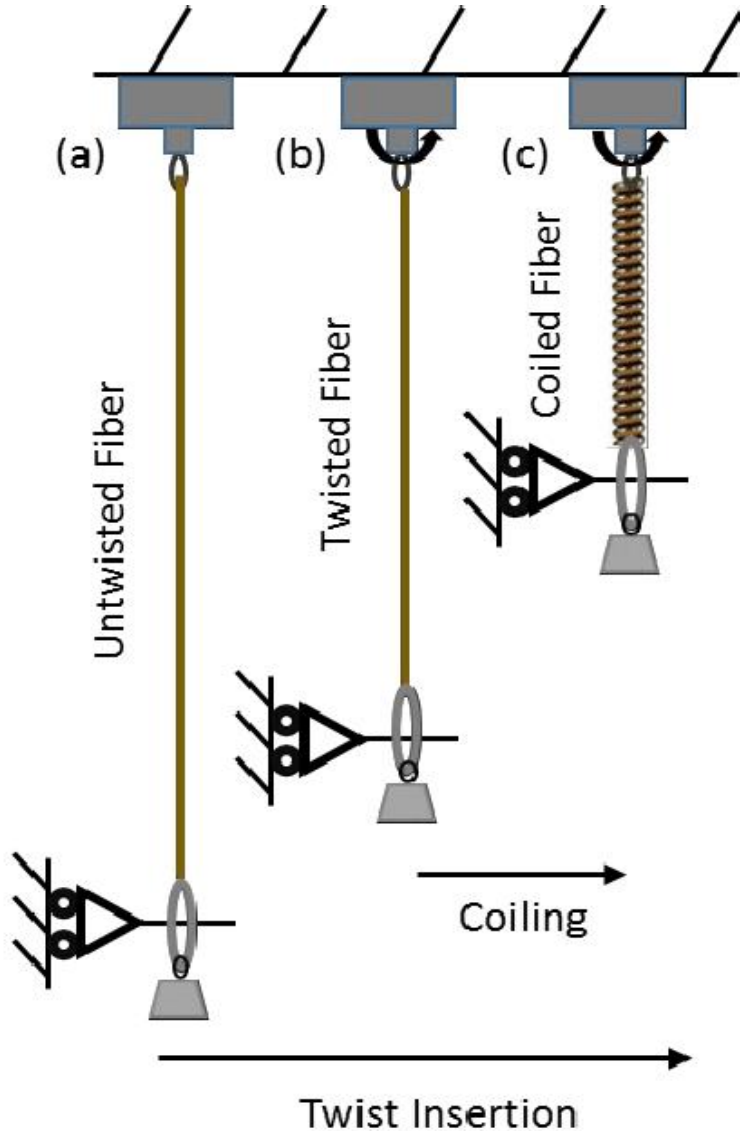
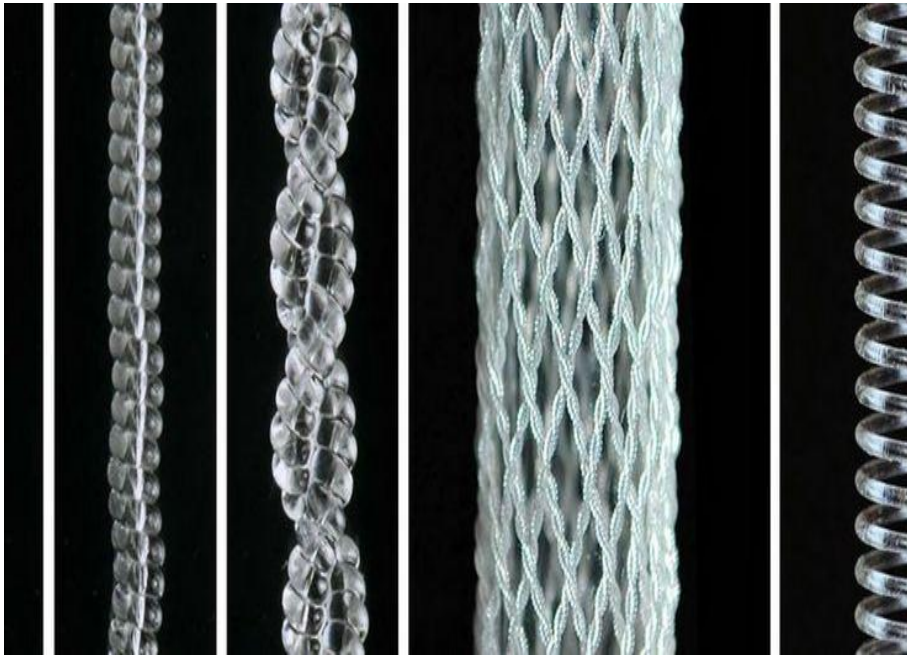
Introduction & Objective

A brief overview of thermal actuator and thre objective of the project

Introduction

Appearance

It appears as a tightly twisted helical spring

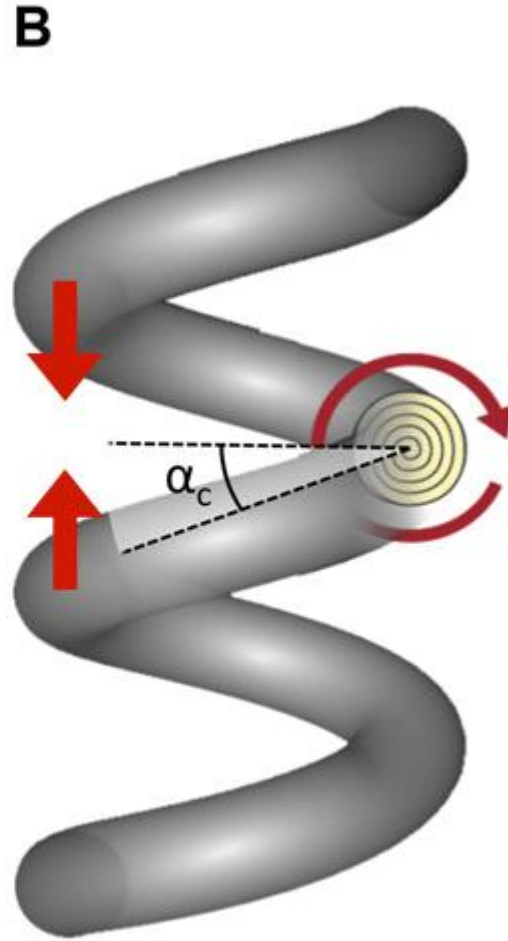
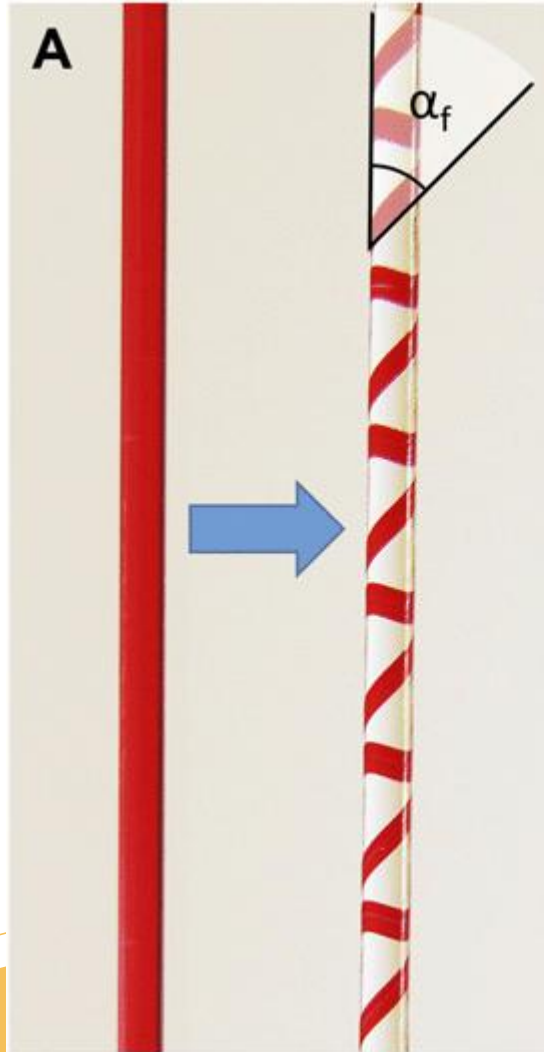


(d)



Introduction

Working Principle



- Apply heat
- Thermal expansion of polymer chains
- Coiled geometry converts expansion into contraction
- Actuator contracts
- Cooling reverses the process

Introduction

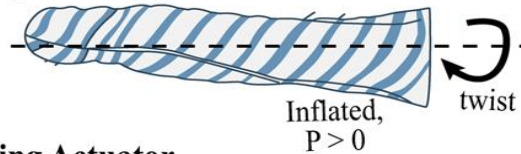
Application

- ✓ Lightweight
- ✓ Low-cost
- ✓ Flexible

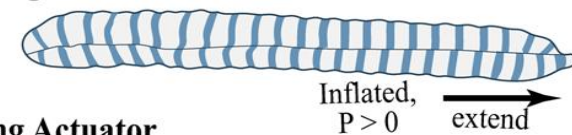
(a) Knit Stretch FRTAs

Collapsed State  Deflated, $P = 0$ 

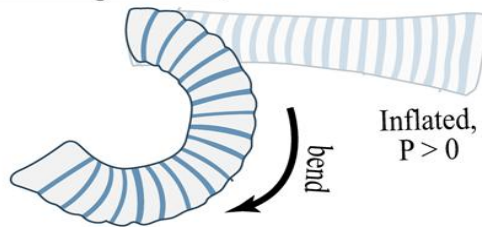
Twisting Actuator



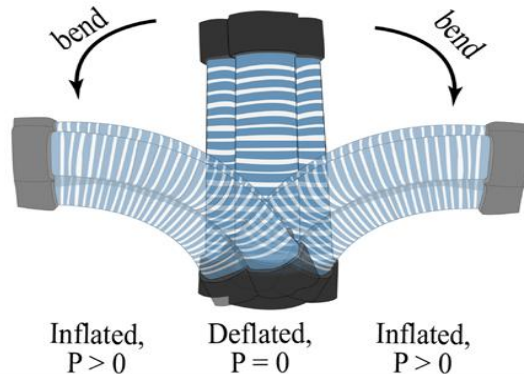
Elongating Actuator



Bending Actuator

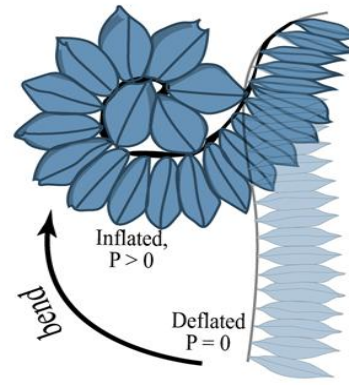


Mutli-DOF Actuator

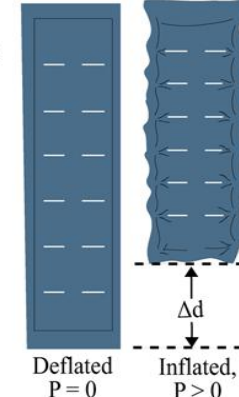


(b) Woven Non-Stretch FSPAs

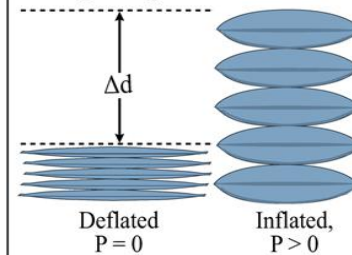
Bending Actuator



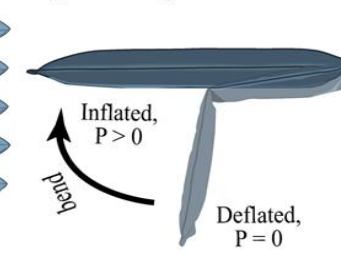
Contracting Actuator



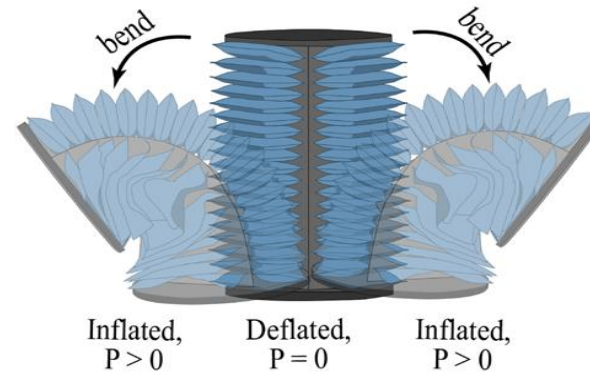
Elongating Actuator



Straightening Actuator

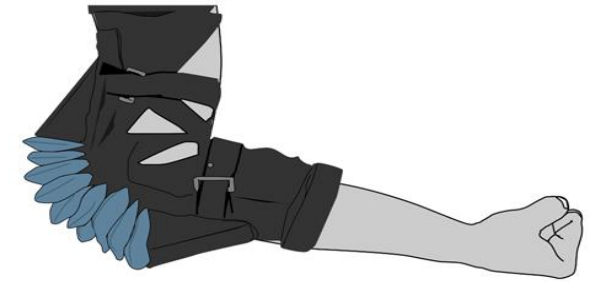


Mutli-DOF Actuator[39]

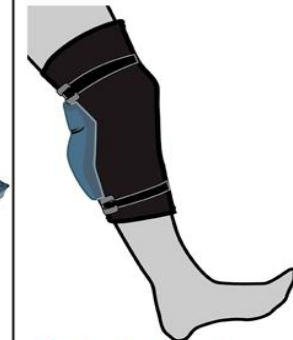


(c) Applications

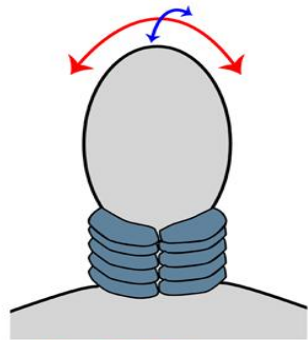
Bicep Curl (*Bending Actuator*)



Knee Extension



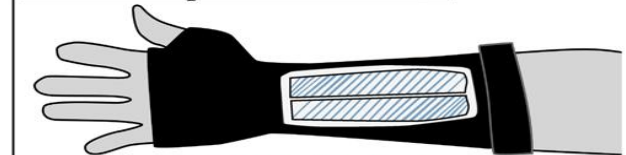
Neck Assitance



Straightening Actuator

Mutli-DOF Actuator

Forearm Supination/Pronation

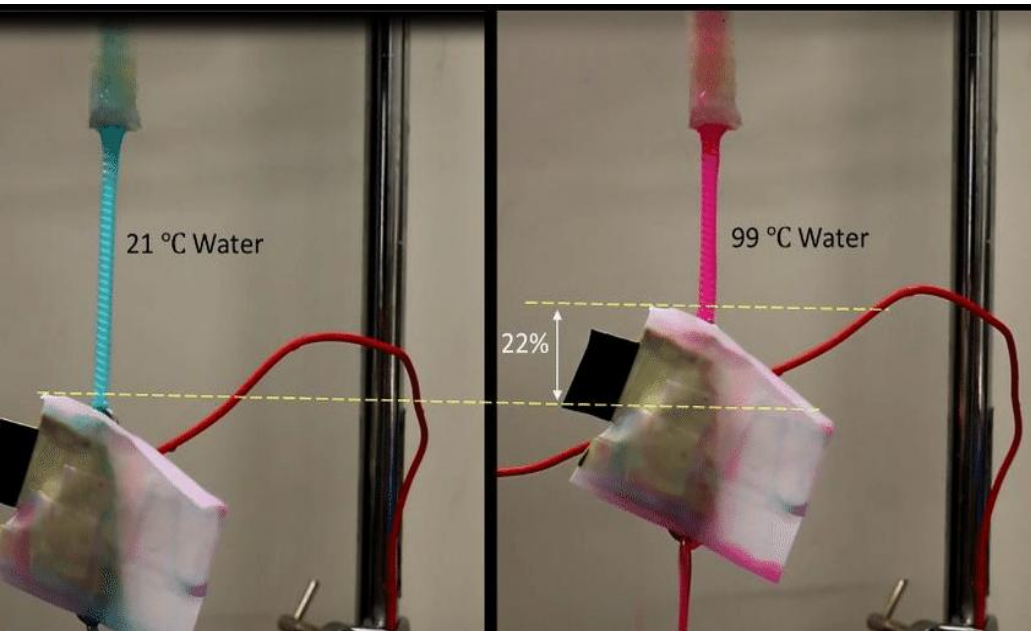


Knit Fabric Twisting Actuator

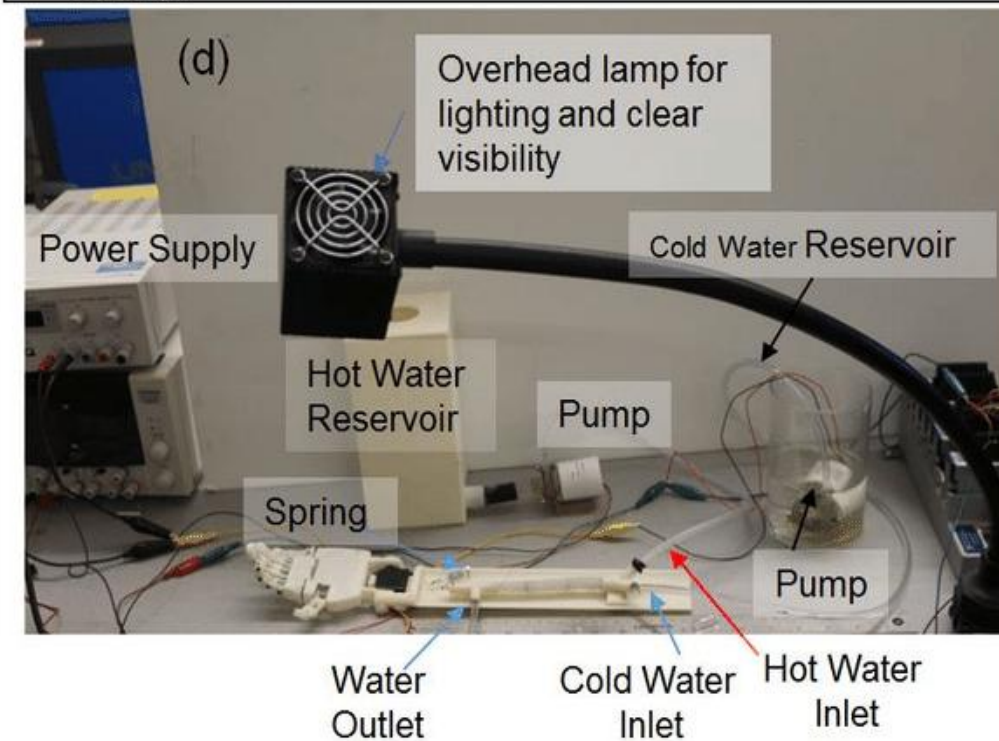
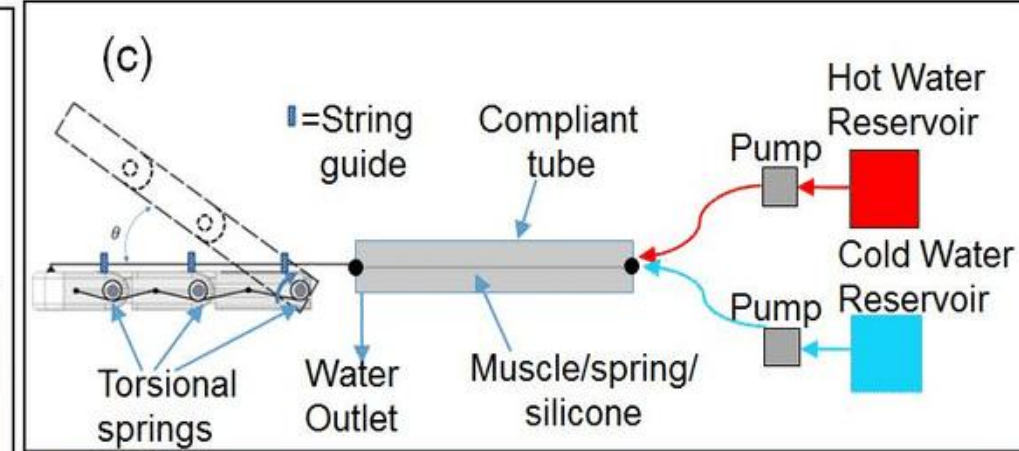
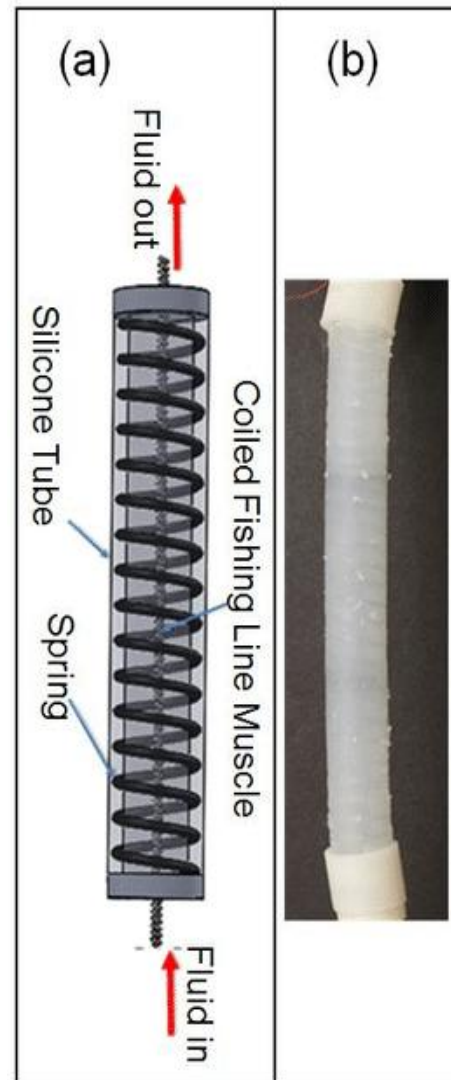


Woven Fabric Contracting Actuator

Introduction



use water temperature difference as power source



Objective

01

Temperature-Actuation Strain Relation

- temperature - strain curve
- optimal temperature

02

Strain-Stress Relation

03

Effect of Diameter and Material on Actuation Performance

- actuation strain for different diameter
- actuation strain for different material

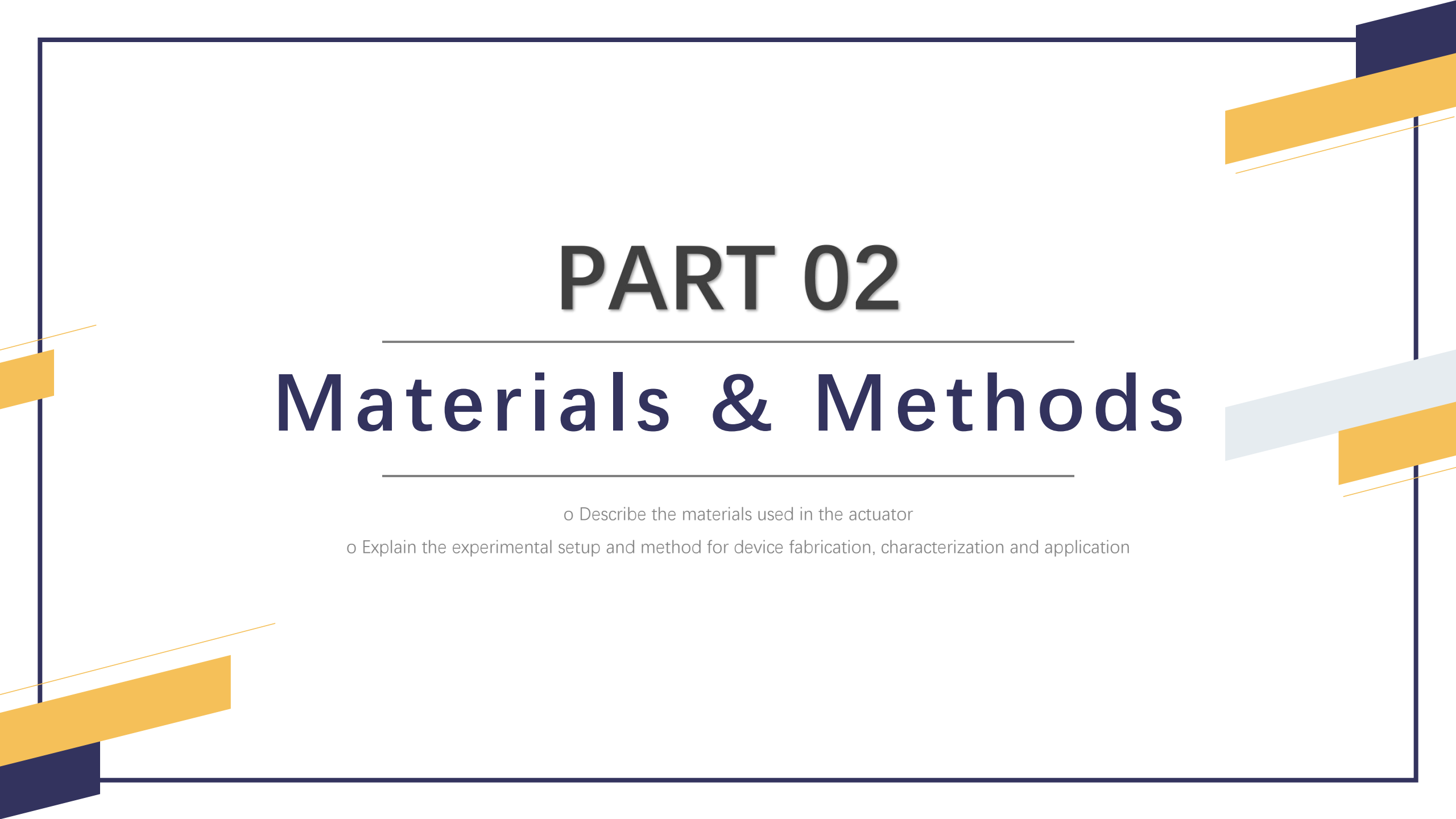
04

Repeatability Test

- repeatability

05

Multi-fibre Actuation Test



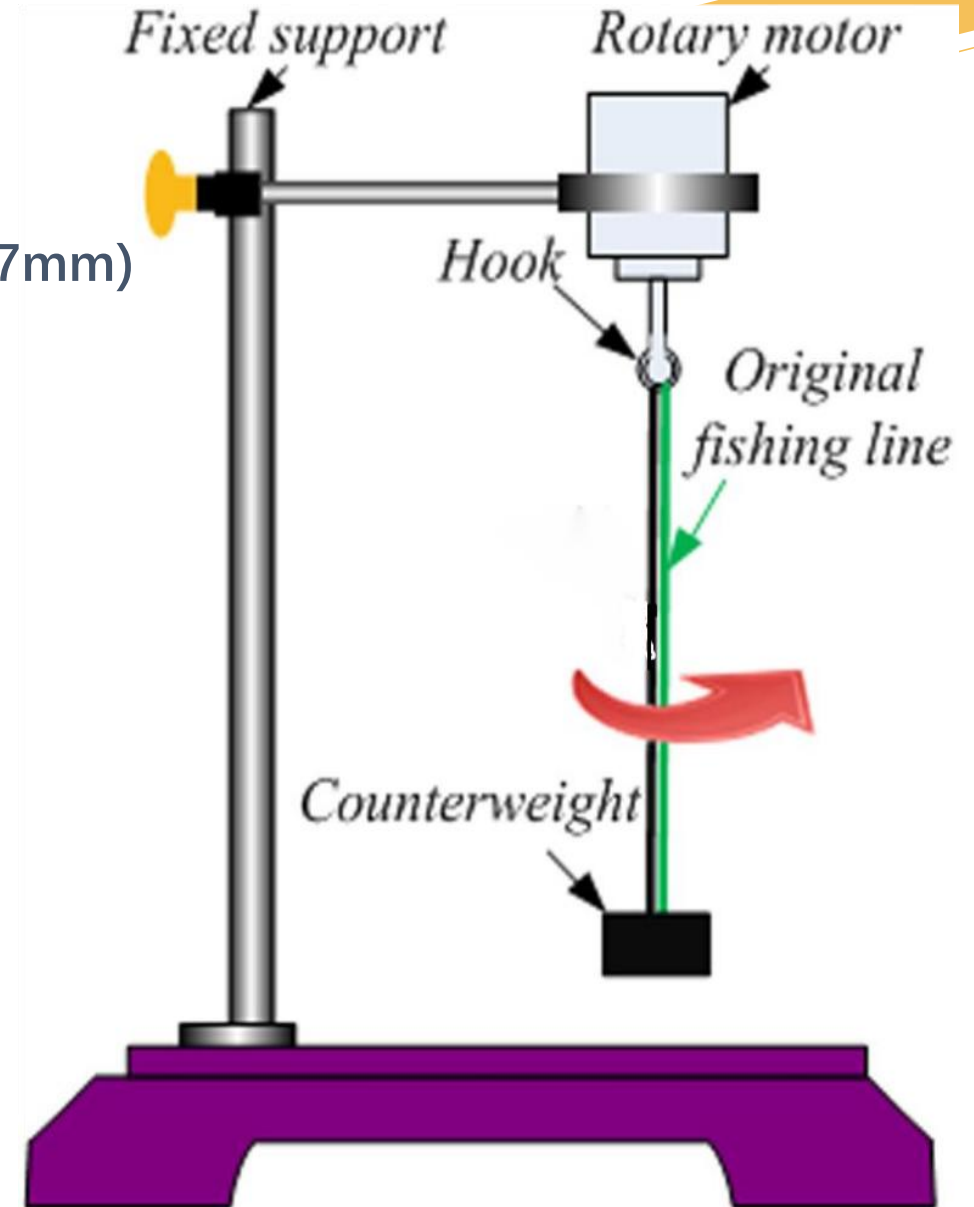
PART 02

Materials & Methods

- o Describe the materials used in the actuator
- o Explain the experimental setup and method for device fabrication, characterization and application

Materials

- Nylon D=0.27mm
- Nylon D=0.45mm
- Polyethylene (PE)(D=0.27mm)
- Twister
- Heat gun
- Phone holder



Methods

01

Fabrication

02

Temperature - Actuation Strain Relation

03

Strain - Stress Relation

04

Frequency Response Experiment

05

Multifibre Actuation Test

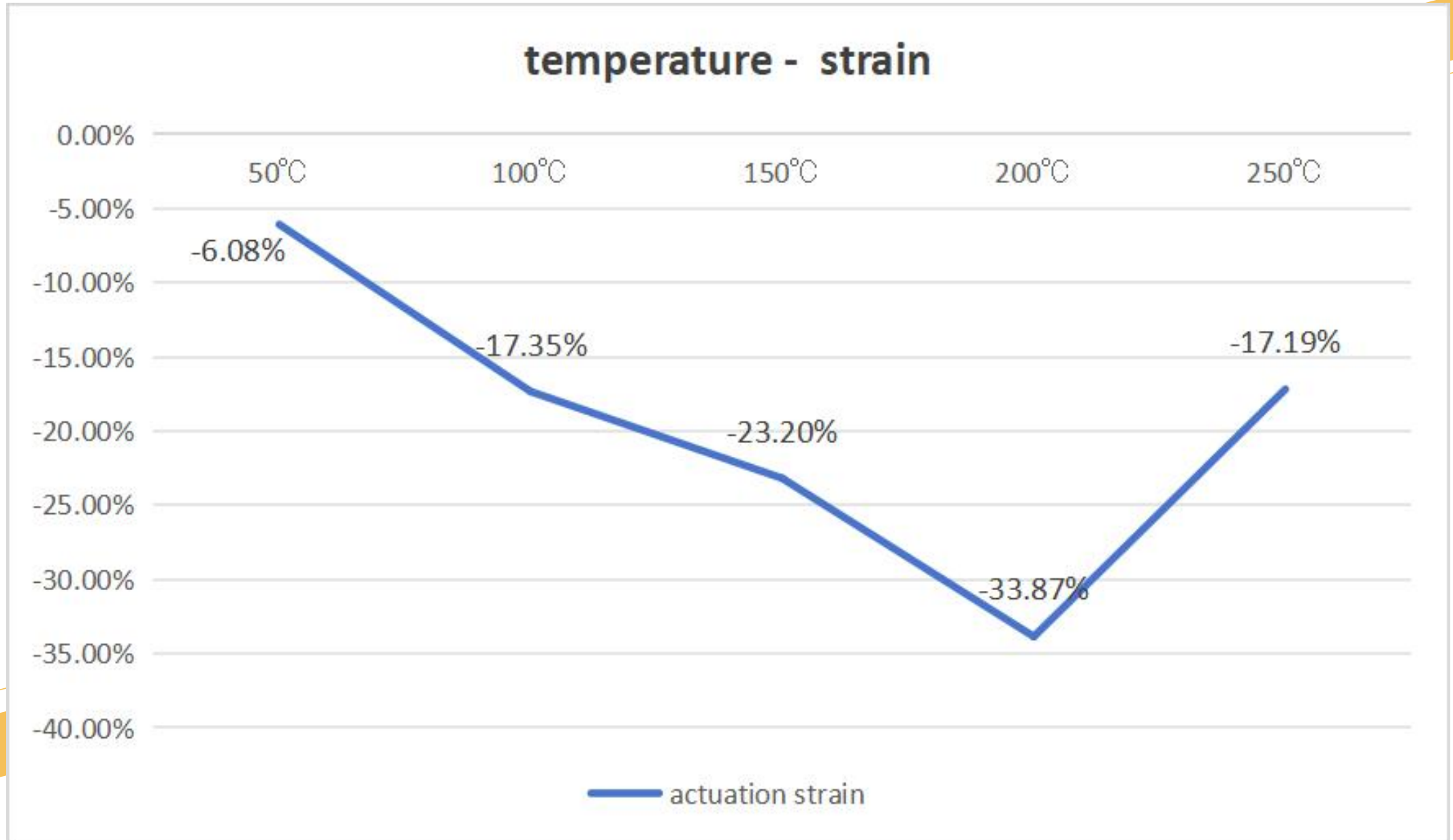


PART 03

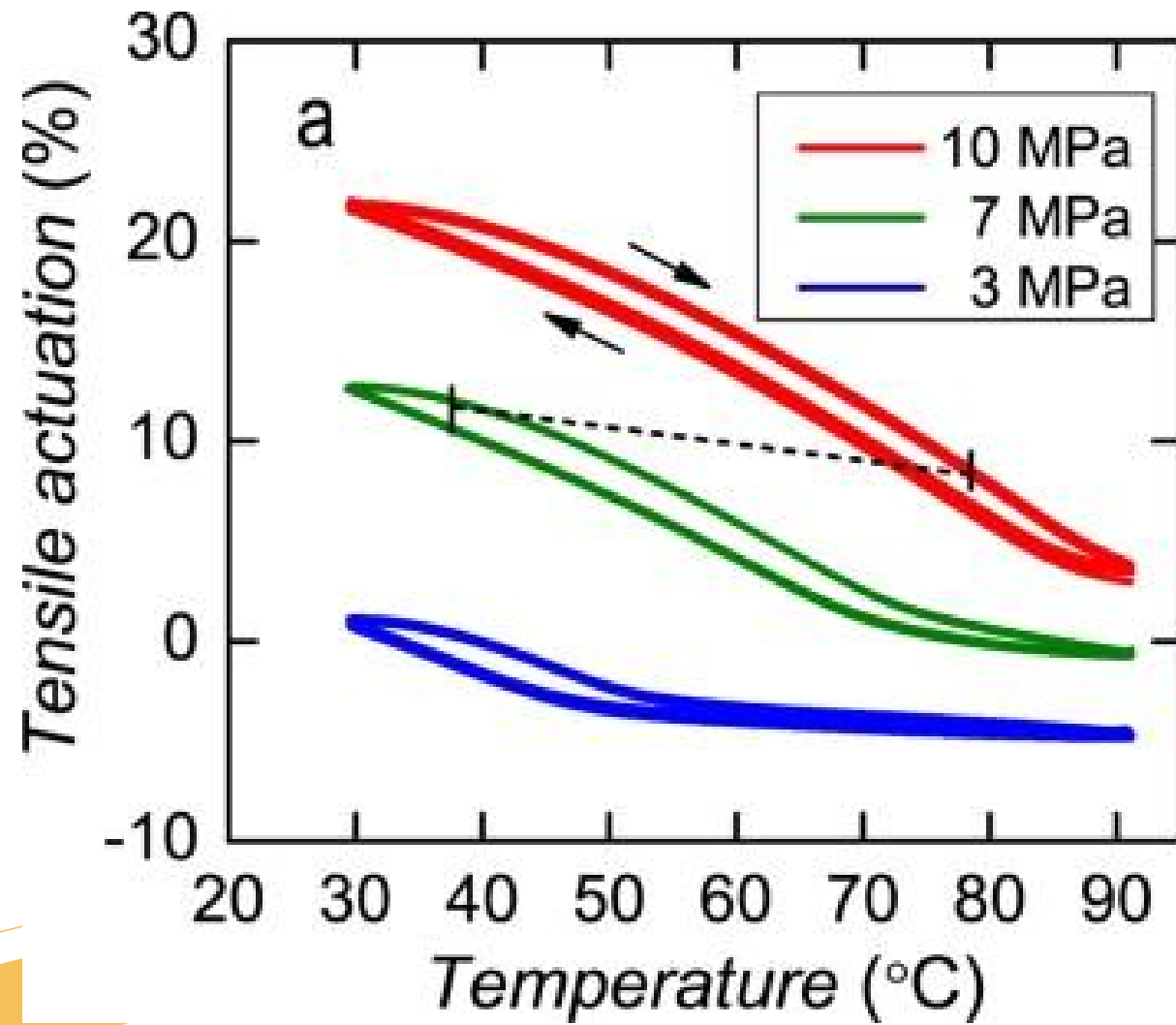
Results & Discussion

- o Present the data from each experiment in a clear and organized format.
- o Include graphs, tables, figures, or videos to visually represent the results.

Results



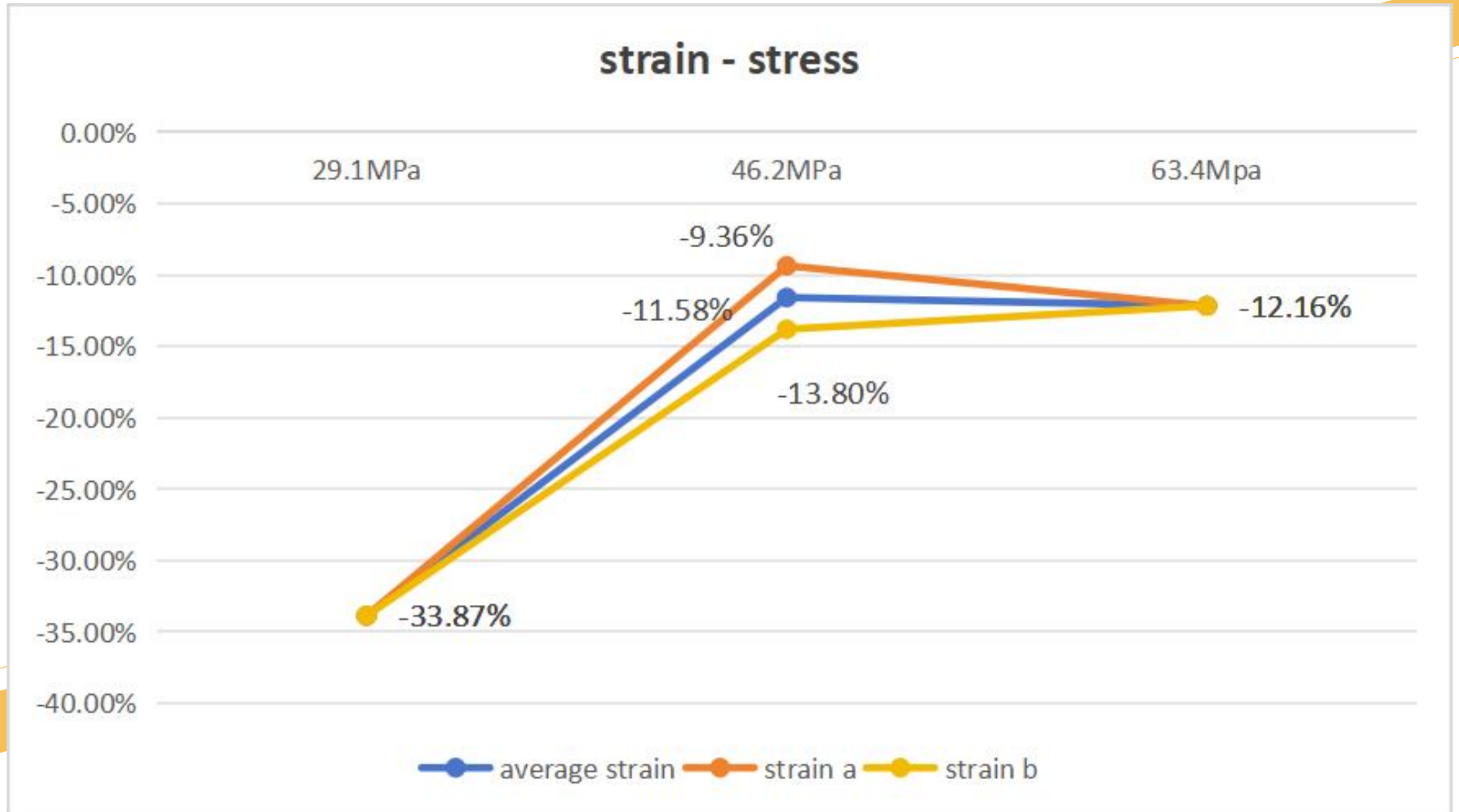
Discussion



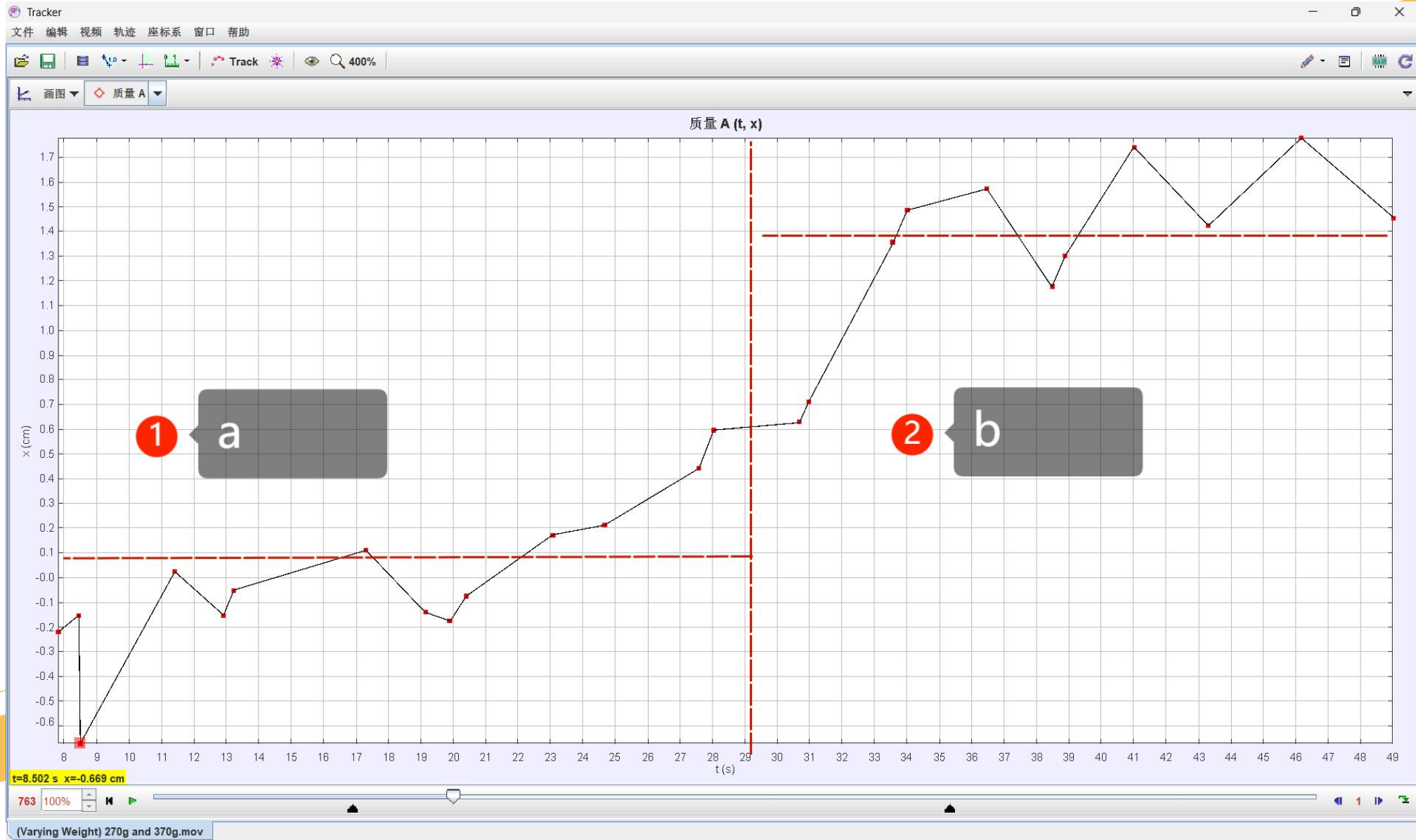
b



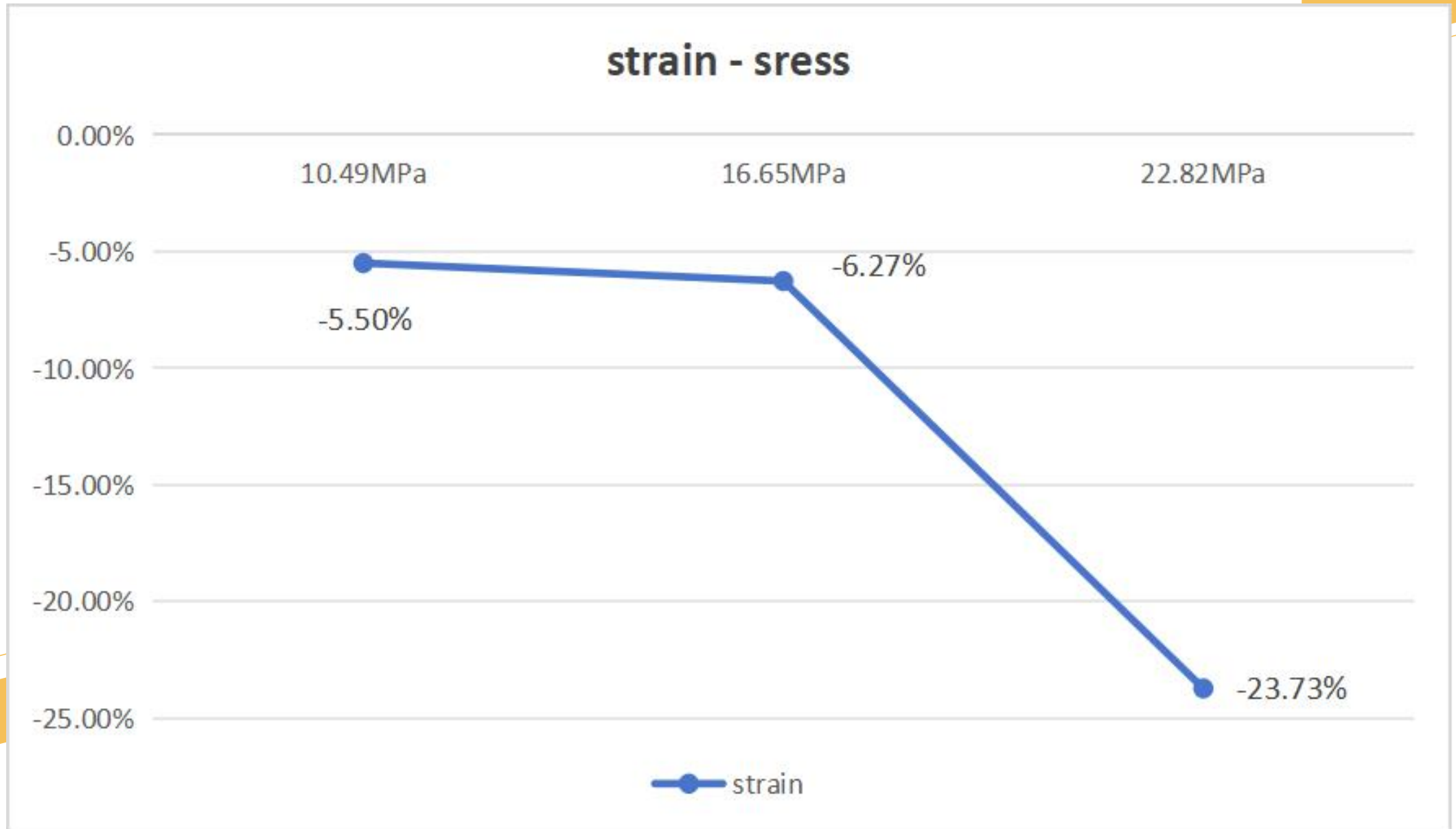
Results



Discussion

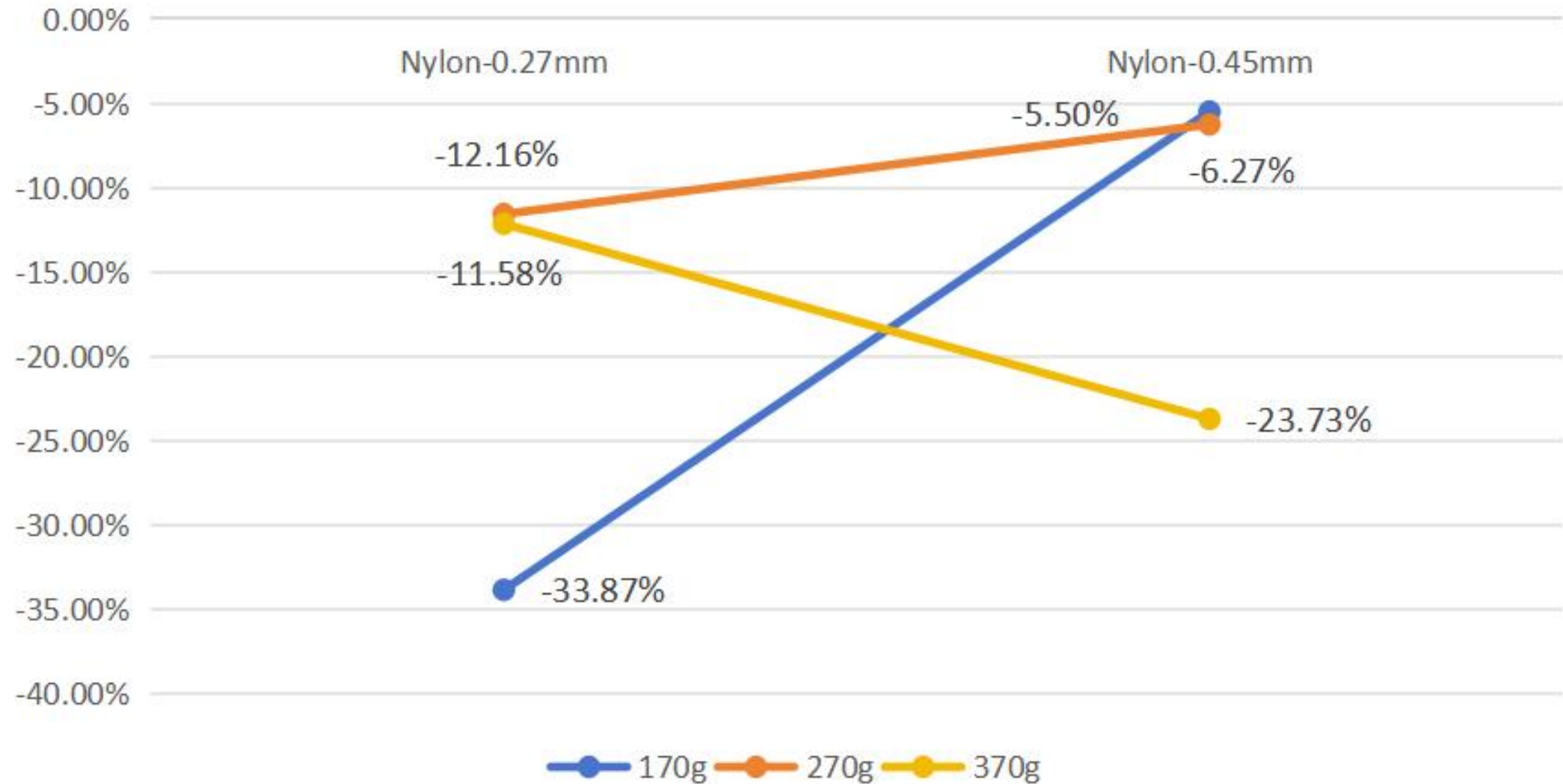


Results



Discussion

strain - diameter



Discussion

Nylon line parameters

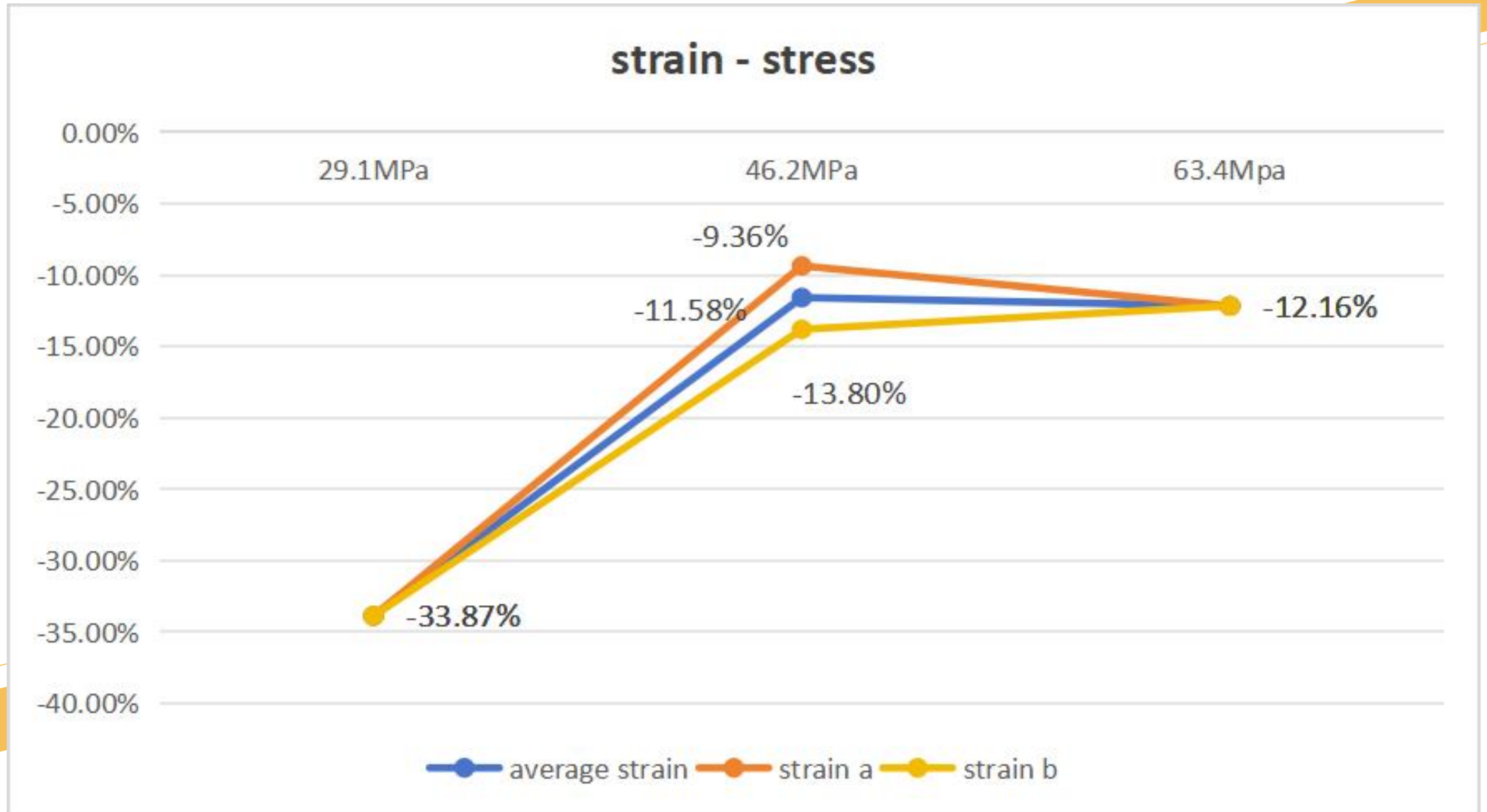
Model	0.4#	0.6#	0.8#	1.0#	1.2#	1.5#	2.0#	2.5#	3.0#	3.5#	4.0#
Diameter MM	0.10	0.12	0.14	0.165	0.18	0.20	0.23	0.26	0.28	0.30	0.32
Pull KG	3.8	4.4	5.1	6.7	7.4	7.9	8.5	9.3	9.9	10.6	12.0

Model	4.5#	5.0#	6.0#	7.0#	8.0#
Diameter MM	0.34	0.37	0.40	0.45	0.50
Pull KG	14.0	16	17.5	18.8	21.0

- Greater stiffness
- Coils are loose
- Carry higher loads without breaking
- Lower strain

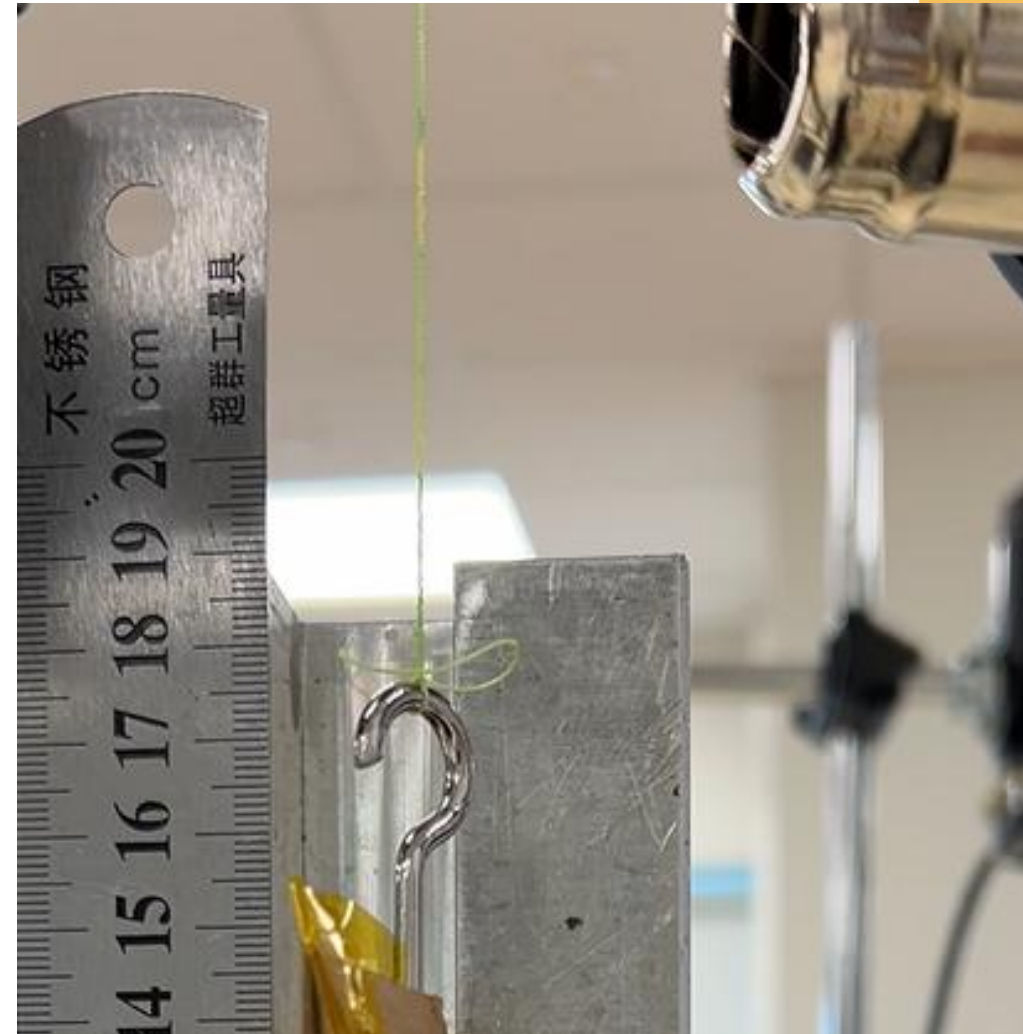


Results

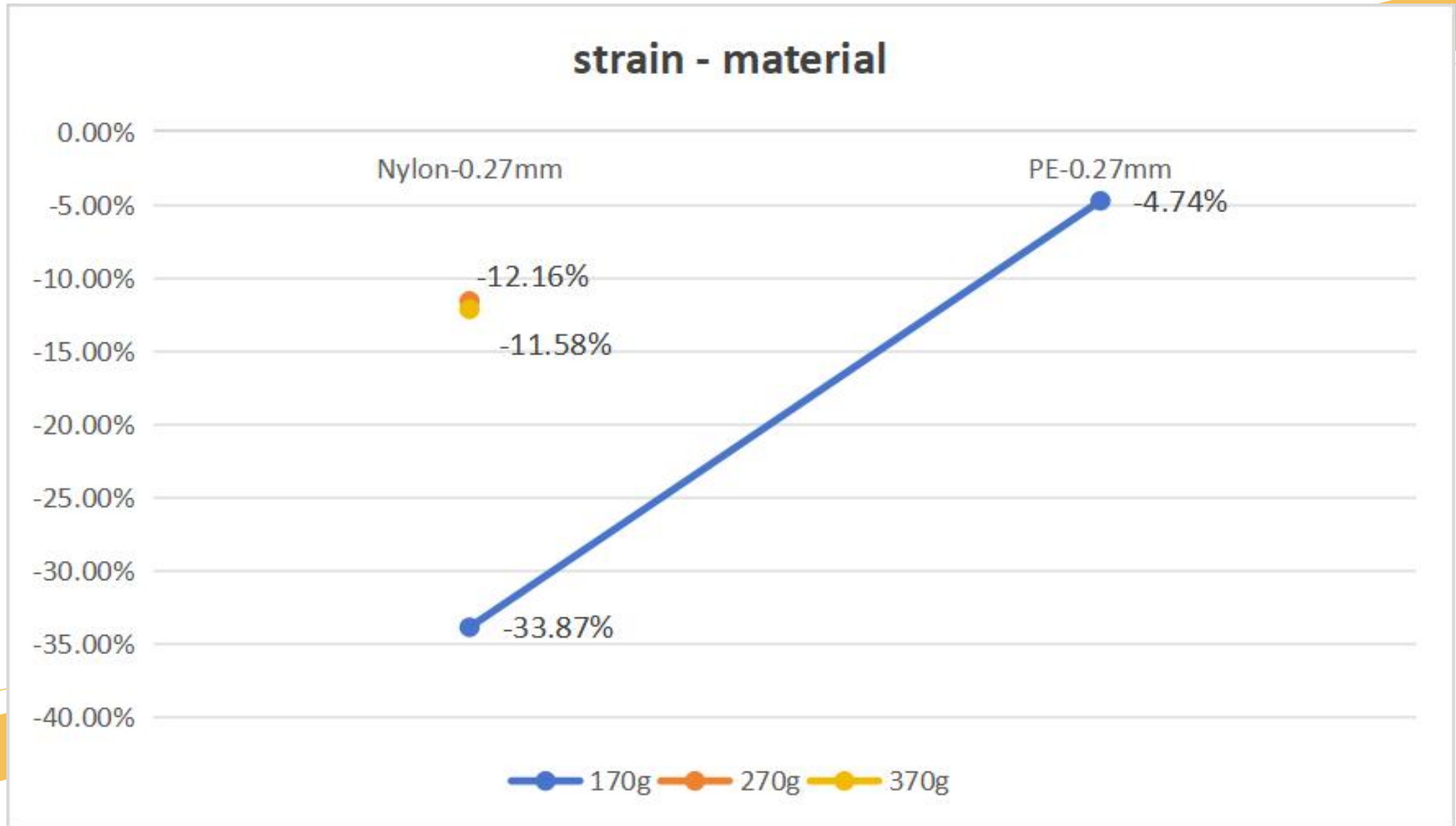


Results

	170g
max displacement (cm)	-0.319
original length (cm)	6.729
strain	-4.74%
Stress (MPa)	29.1

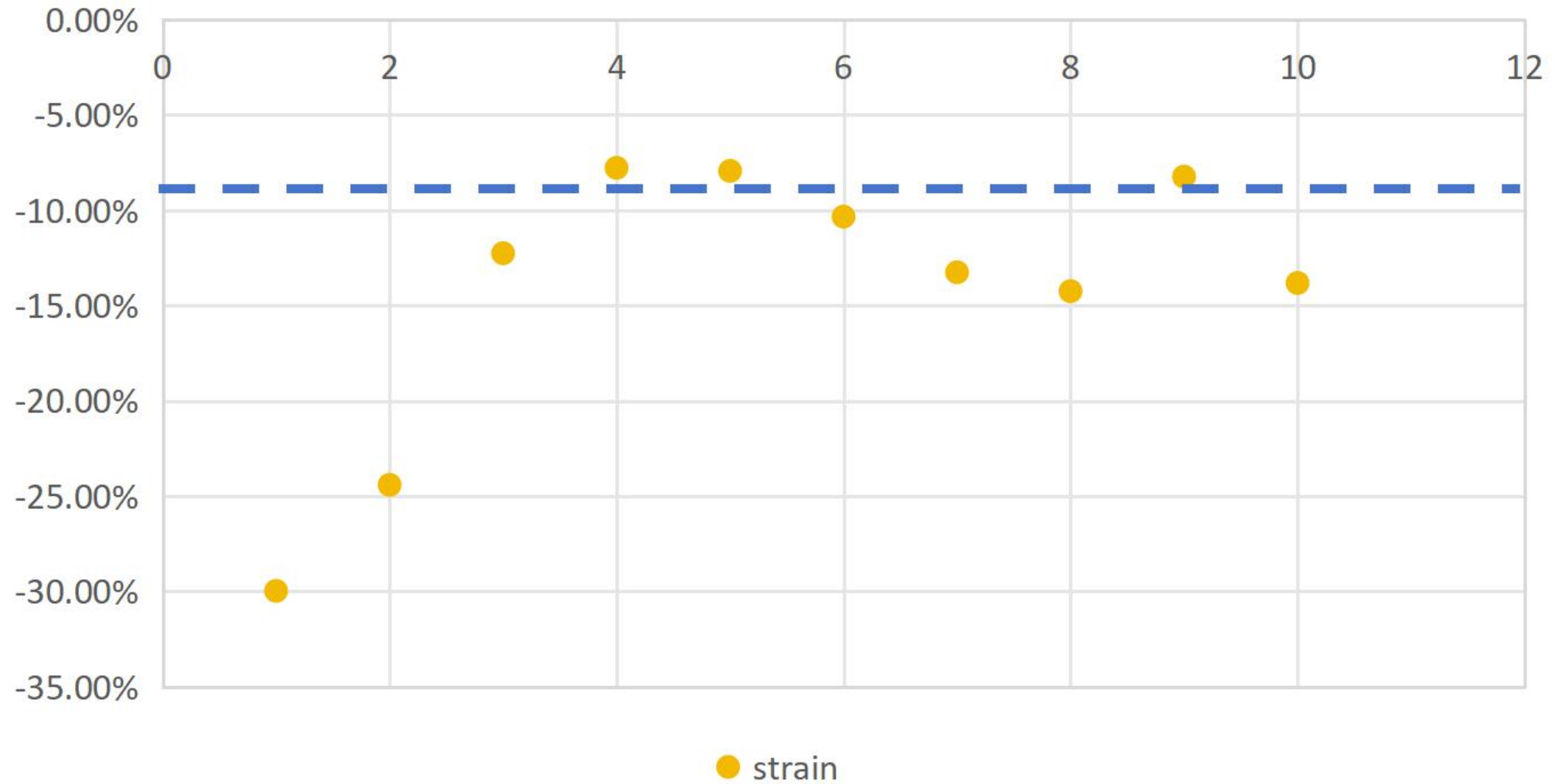


Discussion



Results

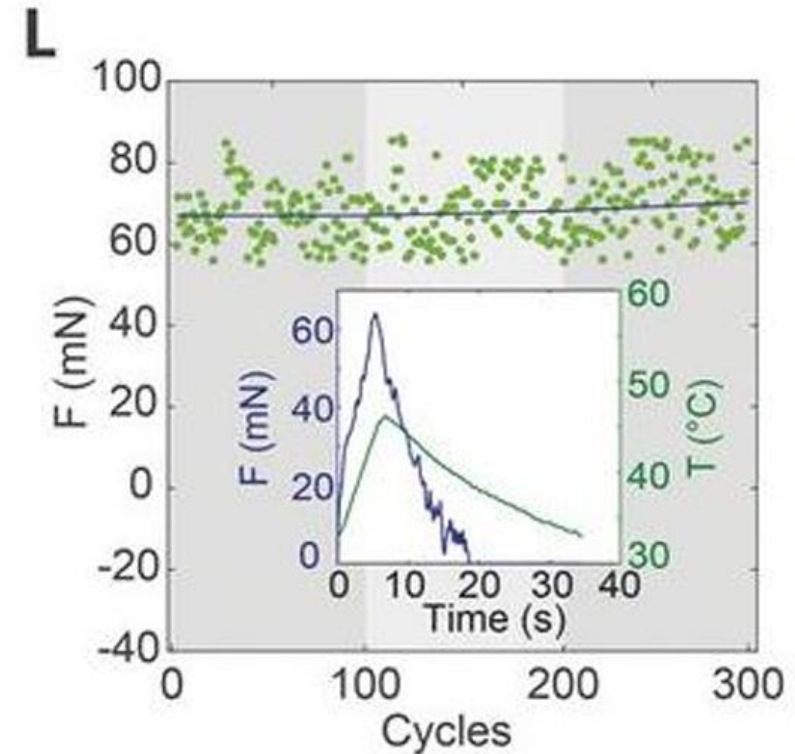
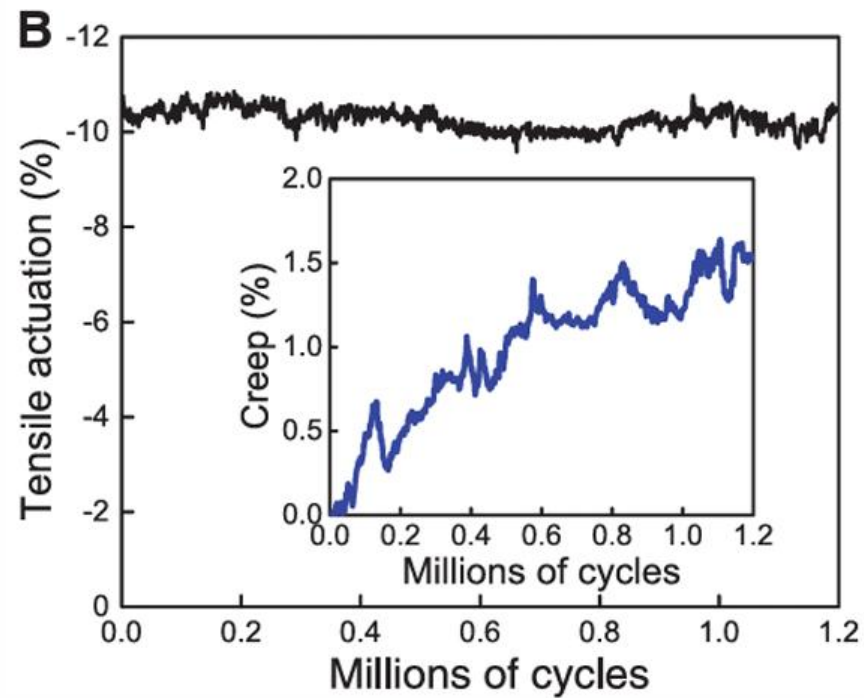
Repeatability



Discussion

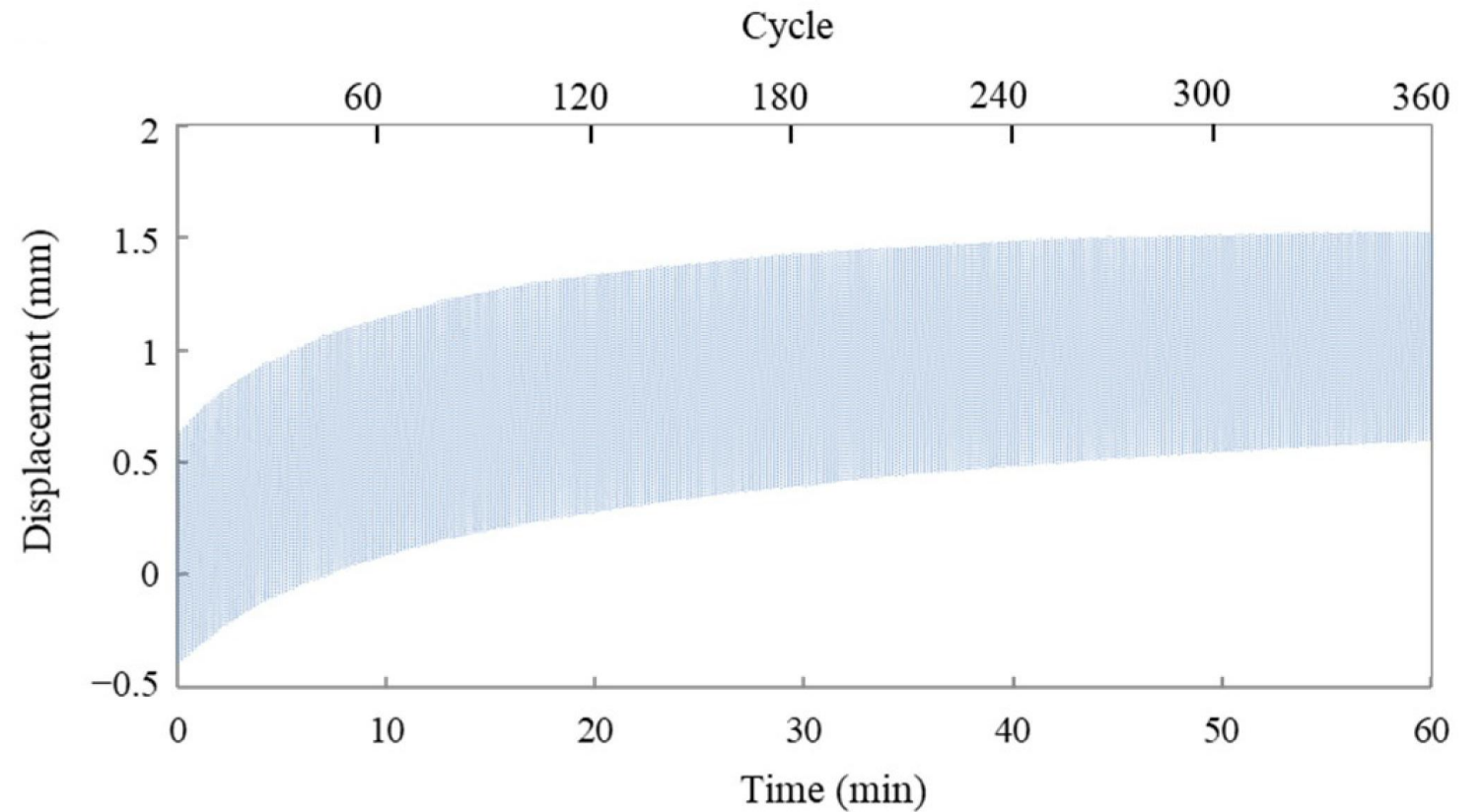
THERMAL ACTUATION

Performance Metric	Pneumatic Polymers	Magnetic Polymers	Dielectric Elastomers	Piezoelectric Polymers	Electrostrictive Polymers	Ionic Polymer–Metal Composites	Thermally Responsive Materials
Repeatability	High	High	High	High	High	Moderate	Moderate



Discussion

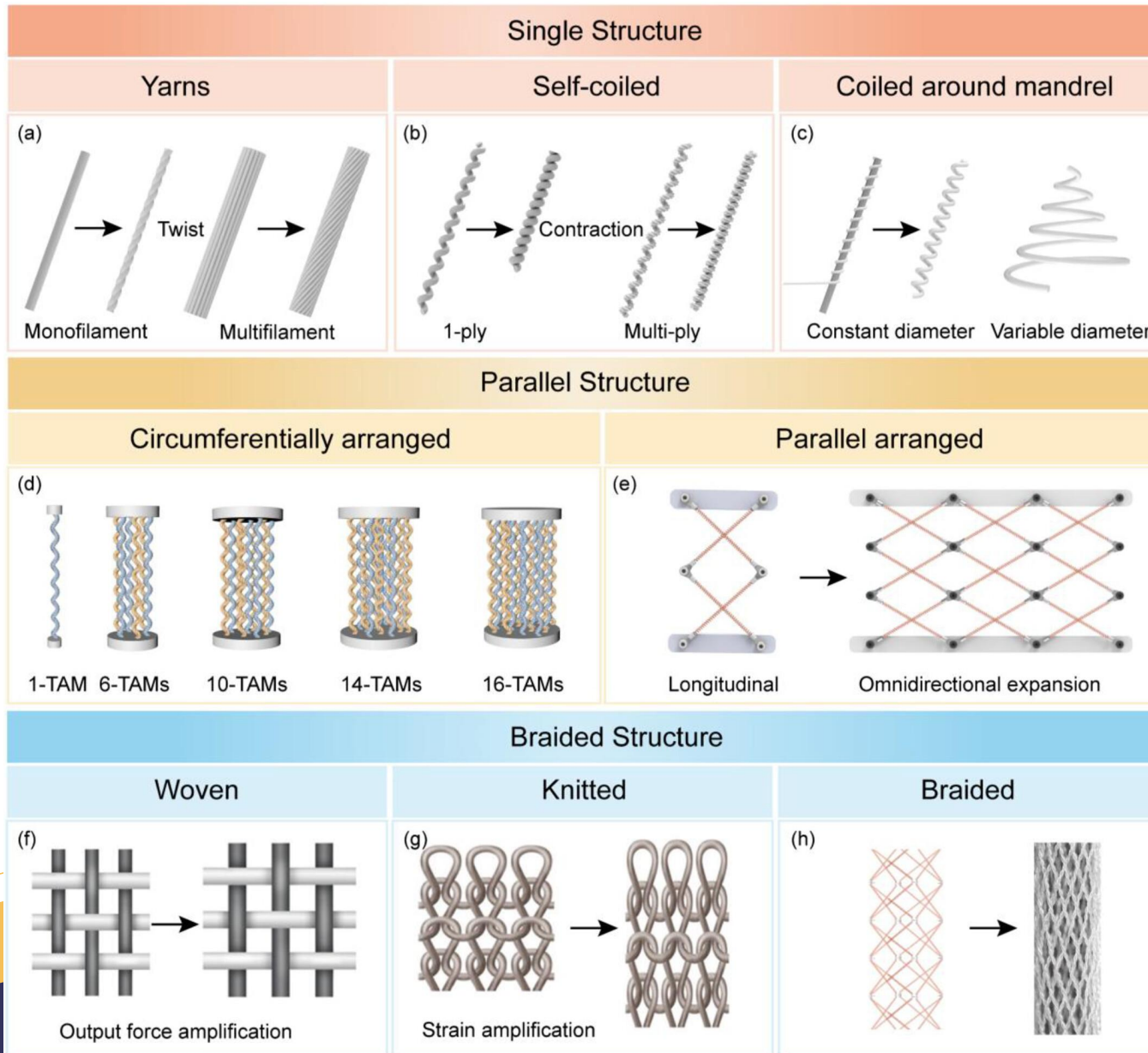
56.00% - 30.29%



Results



Discussion



- Greater stiffness
- Coils are loose
- Carry higher loads without breaking
- Lower strain

Discussion

How to improve the fabrication process



Exceed
Curling



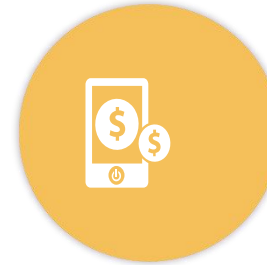
Fast
Heating



Rotaion
Speed



Ueven
Heating



Pre
Stretch

PART 04

Conclusion & References

- o Summarize the key findings.
- o Discuss whether the objectives were addressed.
- o Evaluate the overall success of the experiment.
- o Suggest further improvements

Conclusion

01

**Optimal
Temperature**

02

**Strain Stress
Relation**

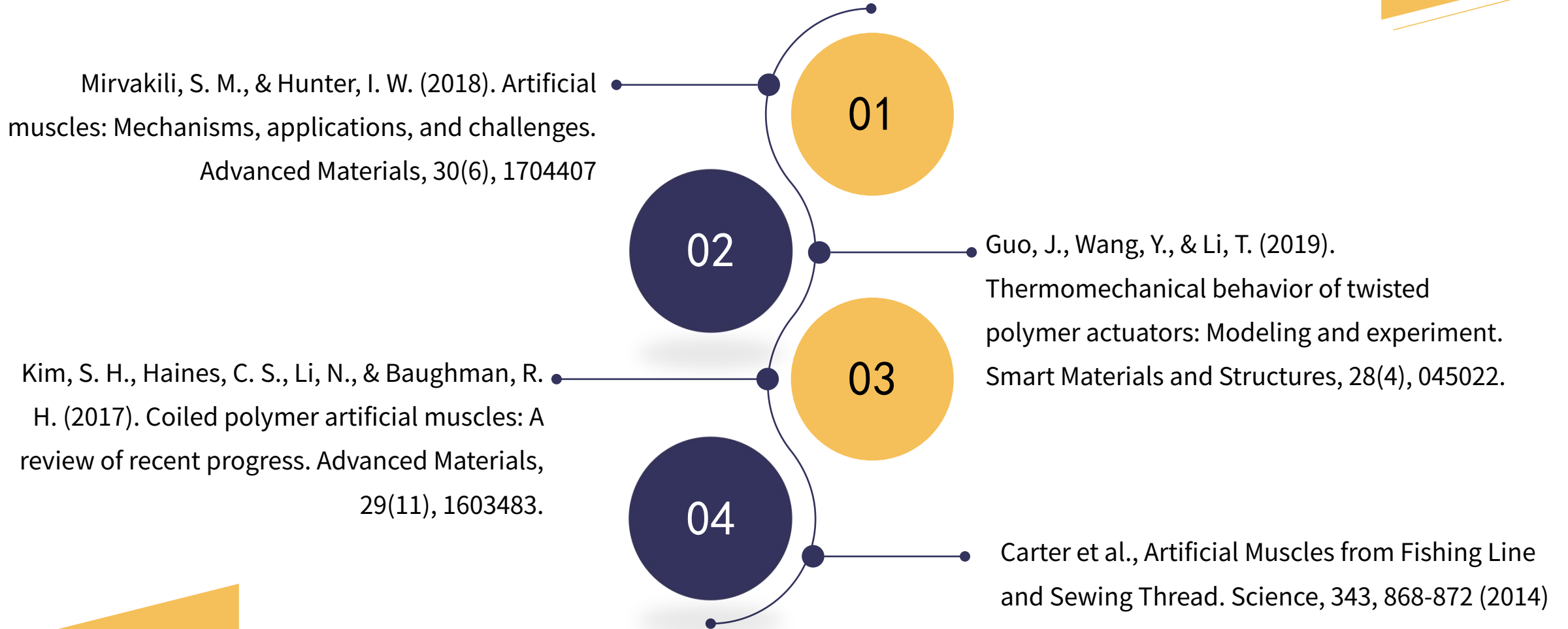
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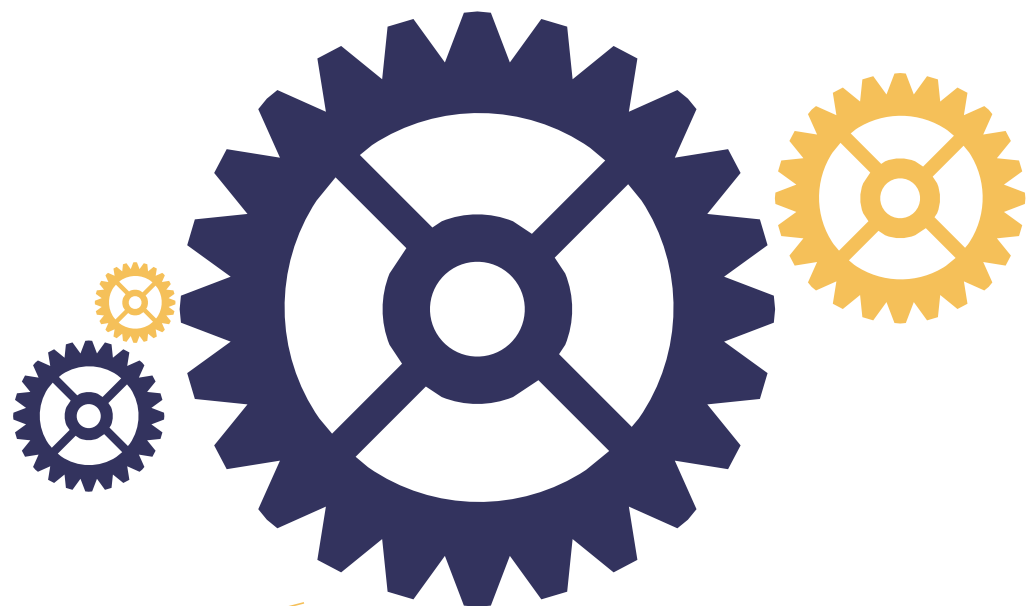
**Repeatability
Test**

04

**Multifiber
Test**

References





Q & A



Thank you

Lu Jianyi